Malaviya National Institute of Technology Jaipur
Department of Computer Science and Engineering

Cryptanalysis of the Vigenère Cipher

Kasiski examination, Index of Coincidence and frequency analysis

Course	Cryptography Laboratory (22CPP307)
Assignment	Assignment 6 (Vigenère Cryptanalysis)
Group	Group 10 (Even group number)
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Instructor	Dr. Meenakshi Tripathi
Submission date	15 October 2026
Assigned ciphertext	Ciphertext-2 (even group numbers) - 758 letters
Recovered key	UNITEDSTATES

1. Objective

Recover the key and plaintext of a Vigenère ciphertext without knowing the key length. The attack estimates the length with the **Kasiski test** (distances between repeated sequences share the key length as a factor), confirms it with the **Index of Coincidence**, splits the text into one Caesar-cipher stream per key position, solves each stream by frequency analysis and verifies the result by re-encryption.

2. Algorithm

```
clean_ciphertext()                  keep A-Z only            -> 758 letters
find_repeated_patterns()             repeated 3..5-grams and their positions
calculate_distances(), find_factors() distances between repeats and their factors
kasiski_analysis()                   rank candidate key lengths by factor counts
calculate_ic()                       average IC of the L groups for L = 1..20
split_into_groups()                  group j = letters j, j+L, j+2L, ...
frequency_analysis(), find_shift()   A-Z counts per group, Caesar shift with minimum chi-square
find_key()                           shifts -> key letters
vigenere_decrypt()                   P = (C - K) mod 26
vigenere_encrypt(), verify()         re-encrypt recovered plaintext, compare with ciphertext
```

3. Results

3.1 Kasiski examination

92 repeated sequences of length 3-5 were found. The factors of their distances (for example 216 = $2^3 \cdot 3^3$, 120 = $2^3 \cdot 3 \cdot 5$) point to the key lengths below; because 12 is divisible by 2, 3, 4 and 6, all of those appear strongly and Kasiski alone cannot decide between them.

Candidate key length	Factor matches
3	90
2	88
6	84
4	83
12	83
8	40
16	26

Candidate key length	Factor matches
13	21

3.2 Index of Coincidence

English text has IC ≈ 0.068; a random polyalphabetic stream has IC ≈ 0.038. Averaging the IC of the groups for each candidate length:

Key length	Average IC	Deviation from 0.068
1	0.0438	0.0242
2	0.0457	0.0223
3	0.0495	0.0185
4	0.0503	0.0177
6	0.0589	0.0091
8	0.0500	0.0180
12	0.0717	0.0037
18	0.0570	0.0110
20	0.0485	0.0195

L = 12 gives IC = 0.0717, essentially the English value, while L = 6 and L = 18 (the divisors/multiples that Kasiski also proposed) only reach 0.059 and 0.057. The key length is therefore **12**.

3.3 Frequency analysis of each group

A-Z FREQUENCY TABLE FOR EACH GROUP (counts of each ciphertext letter)

Grp	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	Top
G1	1	4	1	0	0	3	2	3	6	1	0	4	2	4	3	1	1	0	1	0	6	0	4	3	13	1	Y
G2	8	2	3	0	4	2	6	1	0	0	0	0	0	10	1	1	1	8	2	3	4	3	0	0	3	2	N
G3	3	8	1	1	1	0	2	0	4	0	6	5	6	1	1	4	2	0	0	1	1	5	5	0	0	6	B
G4	2	6	0	0	2	0	3	8	0	0	4	2	11	2	1	0	0	2	0	1	0	0	8	9	2	0	M
G5	1	0	0	0	5	0	1	2	3	1	0	2	6	0	0	4	4	4	9	2	0	5	4	7	2	1	S
G6	0	1	0	7	2	5	2	6	5	1	2	4	0	0	2	2	4	9	0	0	5	1	4	1	0	0	R
G7	4	1	0	3	1	8	0	1	0	4	5	7	1	1	0	0	1	0	5	3	3	2	9	2	1	1	W
G8	4	6	0	0	6	0	0	4	3	0	2	3	8	2	1	0	0	1	0	4	0	1	3	12	1	2	X
G9	6	2	3	4	11	2	1	4	2	0	0	4	0	4	2	1	0	2	3	8	2	0	1	0	1	0	E
G10	6	1	0	0	4	0	5	5	2	0	4	7	6	1	2	2	0	0	0	5	1	1	1	5	2	3	L
G11	0	0	0	0	5	0	1	5	12	0	0	4	5	0	0	4	0	4	8	1	0	2	2	8	2	0	I
G12	3	0	0	2	0	8	7	4	0	2	5	5	1	0	3	0	0	0	3	1	1	3	12	1	0	2	W

Each group is a Caesar cipher. The shift is the one whose decryption has the smallest χ^2 against English letter frequencies:

Group	Size	Shift	Key letter	Min χ^2
1	64	20	U	14.32
2	64	13	N	21.86
3	63	8	I	22.39
4	63	19	T	34.12
5	63	4	E	19.88
6	63	3	D	30.49
7	63	18	S	28.29
8	63	19	T	22.52
9	63	0	A	13.25
10	63	19	T	19.90

Group	Size	Shift	Key letter	Min χ^2
11	63	4	E	19.63
12	63	18	S	24.04

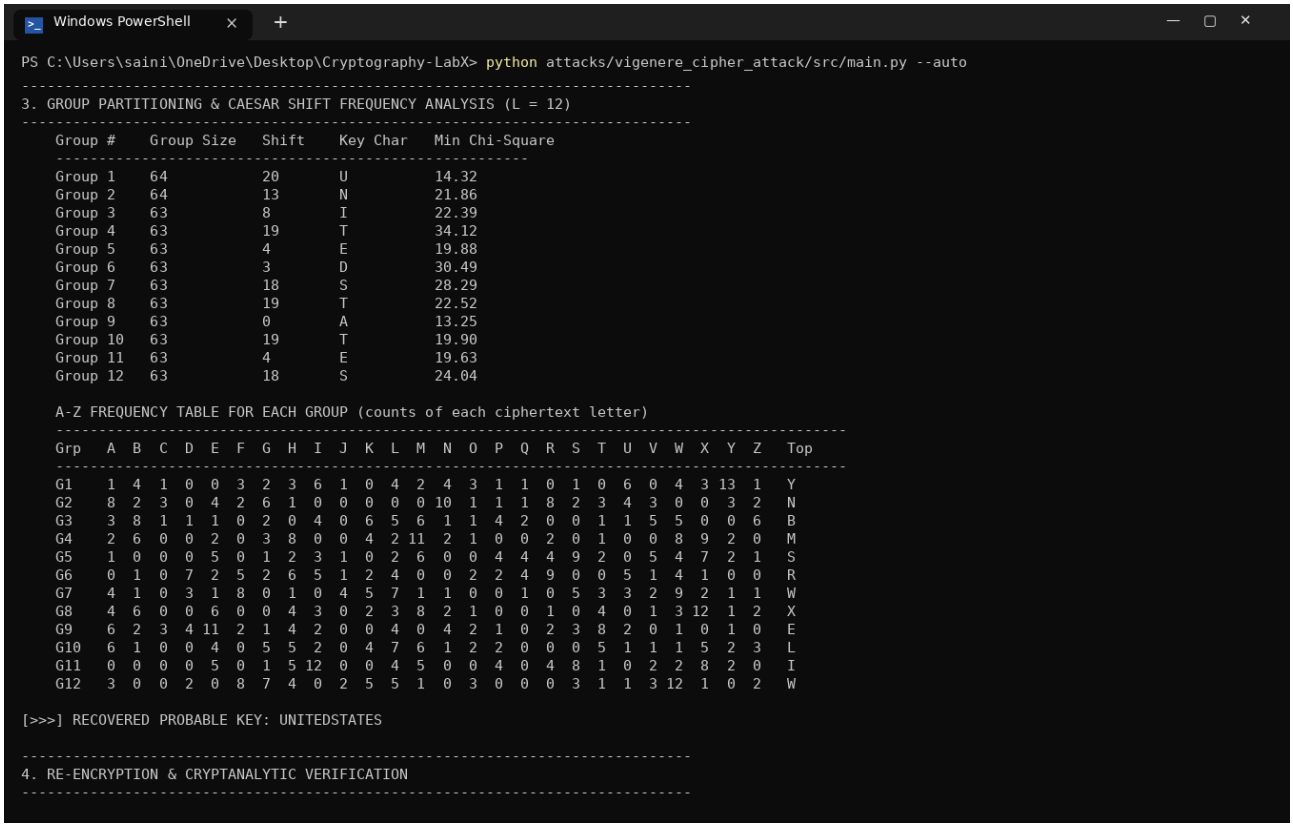


Figure 1 - Group partitioning, frequency tables and recovered key

3.4 Recovered key and plaintext

Key: UNITEDSTATES (the words UNITED STATES). Recovered plaintext (text of the closing resolution of the Declaration of Independence):

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WETHEREFORETHEREPRESENTATIVESOFTHEUNITEDSTATESOFAMERICAINGENERALCONGRE
SSASSEMBLEDAPPEALINGTOTHESUPREMEJUDGEOFTHEWORLDFORTHERECTITUDEOFOURINT
ENTIONSDOINTHENAMEANDBYAUTHORITYOFTHEGOODPEOPLEOFTHESECOLONIESSOLEMNLY
PUBLISHANDDECLARETHATTHESEUNITEDCOLONIESAREANDOFRIGHTOUGHTTOBEFREEANDI
NDEPENDENTSTATESTHATTHEYAREABSOLVEDFROMALLALLEGIANCETOTHEBRITISHCROWNA
NDTHATALLPOLITICALCONNECTIONBETWEENTHEMANDTHESTATEOFGREATBRITAINISANDO
UGHTTOBETOTALLYDISSOLVEDANDTHATASFREEANDINDEPENDENTSTATESTHEYHAVEFULLP
OWERTOLEVYWARCONCLUDEPEACECONTRACTALLIANCESESTABLISHCOMMERCEANDTODOALL
OTHERACTSANDTHINGSWHICHINDEPENDENTSTATESMAYOFRIGHTDOANDFORTHESUPPORTOF
THISDECLARATIONWITHAFIRMRELIANCEONTHEPROTECTIONOFDIVINEPROVIDENCEWEMUT
UALLYPLEDGETOEACHOTHEROURLIVESOURFORTUNESANDOURSACREDHONOR
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3.5 Verification

Re-encrypting the recovered plaintext with the recovered key reproduces the cleaned ciphertext exactly (**100 % match**).

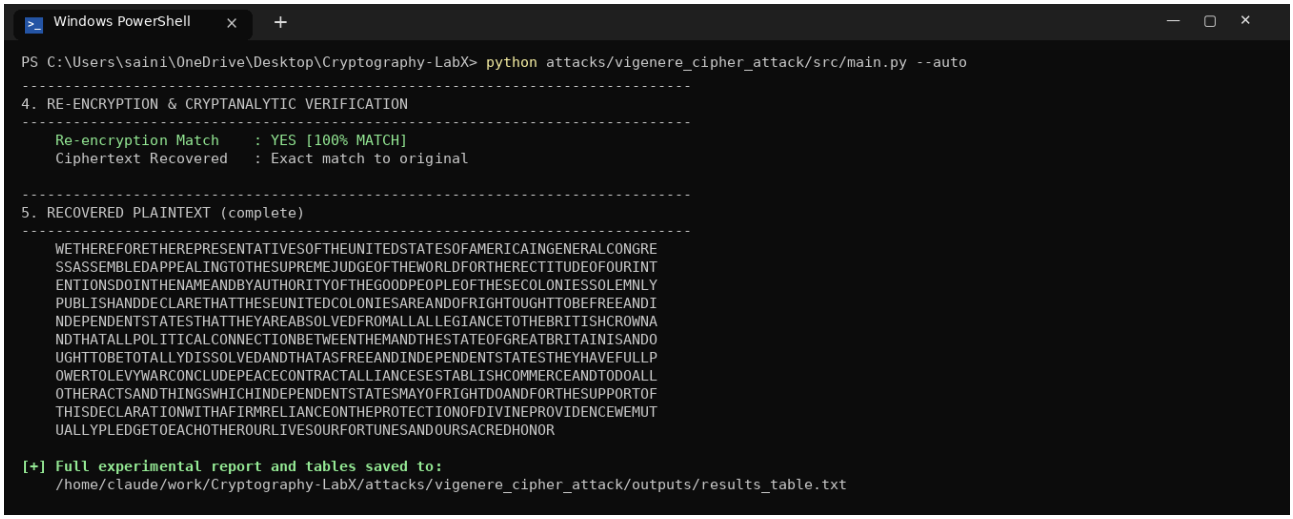


Figure 2 - Re-encryption check and complete plaintext

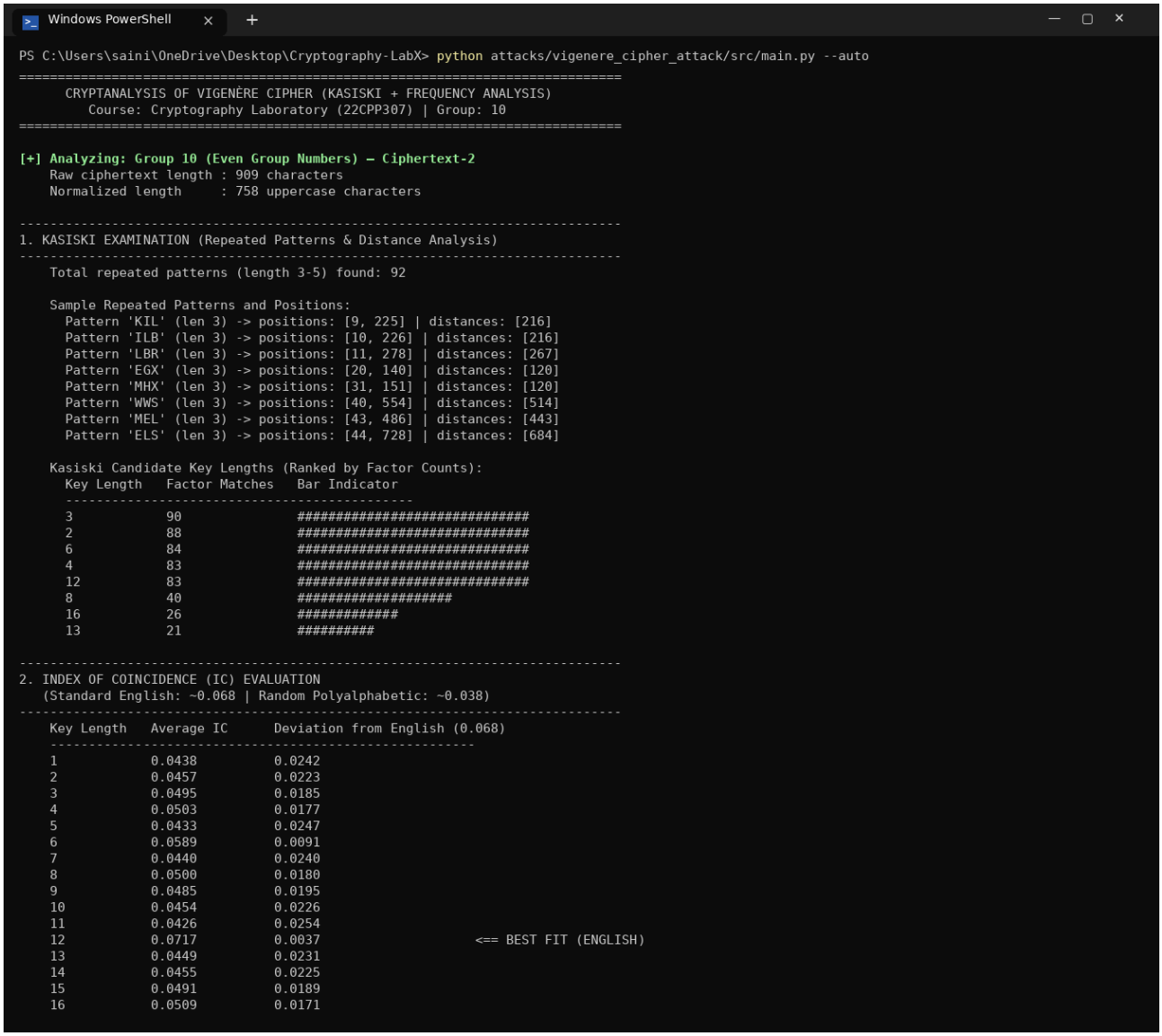


Figure 3 - Kasiski and IC analysis output

4. Discussion

- The Vigenère cipher hides single-letter frequencies only if the key is long and random; a repeating 12-letter key leaks its length through repeated plaintext fragments (Kasiski) and

through the IC.

- Kasiski gave several multiples of the true length; the IC resolved the ambiguity. Using both is more robust than either alone.
- With about 63 letters per group, the minimum-χ^2 shift was correct in every group, confirming that groups of this size are long enough.
- A natural-language key (UNITEDSTATES) would also fall to a dictionary attack, but a short random key of the same length would still be broken by the method above.

5. Conclusion

The key length (12), the key (UNITEDSTATES) and the full plaintext were recovered from the ciphertext alone and validated by re-encryption.